Experimental Evaluation and Hardware Overhead Analysis of Zero-Blocking Optimization in a 4x4 Tensor Grid Accelerator

Objective

To quantitatively evaluate the operation of the *zero-Blocking circuit* in a 4×4 matrix multiplication accelerator by measuring the circuit's switching activity as a function of input data sparsity.

Methodology & Experimental Setup

- **Simulation Tools:** Icarus Verilog for compilation/execution, coupled with a custom C-based **VPI (Verilog Procedural Interface)** monitoring module (activity.c) registered directly into the simulation runtime.
- **Evaluation Metric:** The total number of signal value changes monitored strictly at the logic level during active instruction execution. This provides an exact indicator of dynamic power consumption.
- **VPI Activity Isolation:** The VPI C-module isolates toggles per test execution step, eliminating testbench noise floor overhead and yielding true hardware-level switching metrics.
- **Test Scenarios:** Executed via zero_block_eval_tb.v (make eval), consisting of a dense baseline workload followed by progressive tile-level zero-blocking evaluations (0 to 3 zeroed-out 2×2 Tensor Tiles, corresponding to 0% to 75% matrix sparsity).

Execution Trace Snippet

```
VCD info: dumpfile tb_tensor.vcd opened for output.
SWITCHING ACTIVITY: 497 Toggles
TEST 1: Signed Tensor MAC Result (A x B + C = D)
----------------------------------------
     ┌   2   -1    1    0┐
  A =│  -2    3    2   -5│
     │  -4    1    1    1│
     └   0    2   -1    2┘
----------------------------------------
     ┌   1    2   -2    1┐
  B =│   3   -1    0    2│
     │   2    0    1    2│
     └   1   -4    3    1┘
----------------------------------------
     ┌ -10   10    0    3┐
  C =│   5   -2    4    1│
     │ -16    2    2   -6│
     └   1    5    3    0┘
----------------------------------------
     ┌  -9   15   -3    5┐
  D =│  11   11   -5    4│
     │ -14  -11   14   -5│
     └   7   -5    8    4┘
----------------------------------------
SWITCHING ACTIVITY: 1106 Toggles
TEST 2: Zero Blocking ON - Matrix A with 1 Zero-Tile (A00)
----------------------------------------
     ┌   0    0    2    1┐
  A =│   0    0    3   -2│
     │   1    4   -3    2│
     └   2    1    1    3┘
----------------------------------------
     ┌   2    1    1    2┐
  B =│  -2    3    0    1│
     │   3   -3    1    0│
     └   1    2   -2    1┘
----------------------------------------
```

```
     ┌   0    0    0    0┐
  C =│   0    0    0    0│
     │   0    0    0    0│
     └   0    0    0    0┘
----------------------------------------
     ┌   7   -4    0    1┐
  D =│   7  -13    7   -2│
     │ -13   26   -6    8│
     └   8    8   -3    8┘
----------------------------------------
SWITCHING ACTIVITY: 1746 Toggles
TEST 3: Zero Blocking ON - Matrix A with 2 Zero-Tiles (A00, A01)
----------------------------------------
     ┌   0    0    0    0┐
  A =│   0    0    0    0│
     │   2    1    1    2│
     └  -2    3    4   -3┘
----------------------------------------
     ┌   2    1    1    2┐
  B =│  -2    3    0    1│
     │   3   -3    1    0│
     └   1    2   -2    1┘
----------------------------------------
     ┌   0    0    0    0┐
  C =│   0    0    0    0│
     │   0    0    0    0│
     └   0    0    0    0┘
----------------------------------------
     ┌   0    0    0    0┐
  D =│   0    0    0    0│
     │   7    6   -1    7│
     └  -1  -11    8   -4┘
----------------------------------------
SWITCHING ACTIVITY: 2149 Toggles
TEST 4: Zero Blocking ON - Matrix A with 3 Zero-Tiles (A00, A01, A10)
----------------------------------------
     ┌   0    0    0    0┐
  A =│   0    0    0    0│
     │   0    0    3    2│
     └   0    0   -2    1┘
----------------------------------------
     ┌   2    1    1    2┐
  B =│  -2    3    0    1│
     │   3   -3    1    0│
     └   1    2   -2    1┘
----------------------------------------
     ┌   0    0    0    0┐
  C =│   0    0    0    0│
     │   0    0    0    0│
     └   0    0    0    0┘
----------------------------------------
     ┌   0    0    0    0┐
  D =│   0    0    0    0│
     │  11   -5   -1    2│
     └  -5    8   -4    1┘
----------------------------------------
SWITCHING ACTIVITY: 2484 Toggles
zero_block_eval_tb.v:280: $finish called at 195 (1s)
```

Analysis & Benchmark Results

By decoupling cumulative activity measurements, the isolated execution cost (ΔToggles) for each individual workload is extracted:

Test Scenario	Sparsity Level	Zero-Blocking Status	Cumulative Toggles	Execution Cost	Dynamic Power Reduction / Dense Baseline
Initial Reset & Setup	-	OFF	497	497	-
TEST 1 (Dense Matrices)	0%	OFF	1,106	**609**	**0.0%**
TEST 2 (1 Zero Tile - A_{00})	25%	ON	1,746	**640**	**+5.1%**
TEST 3 (2 Zero Tiles - A_{00}, A_{01})	50%	ON	2,149	**403**	**-33.8%**
TEST 4 (3 Zero Tiles - A_{00}, A_{01}, A_{10})	75%	ON	2,484	**335**	**-45.0%**

Technical Observations

1. **Isolation of Real Hardware Gains:** An in-memory VPI event monitor reveals that the Zero-Blocking optimization delivers up to **45.0% dynamic switching activity reduction** at 75% sparsity.
2. **Control Logic Overhead at Low Sparsity:** At 25% sparsity (1 Zero Tile), execution cost increases slightly from 609 to 640 toggles (+5.1%). This reflects the fixed activation overhead of the zero-detection comparators and gating multiplexers.
3. **High Sparsity Scaling:** Once sparsity reaches $\geq 50\%$, the power saved by gating entire multiplier-accumulator (MAC) sub-trees far outweighs the detection logic cost, reducing execution activity down to **335 toggles** at 75% sparsity.

Conclusions

- **High Efficiency in Sparse Workloads:** The Zero-Blocking mechanism is highly effective, delivering up to a **45% reduction in dynamic switching activity** during matrix multiplication.
- **Architectural Trade-Off:** While single-tile gating exhibits a minor control logic overhead (~5%), the architecture provides steep power savings for medium-to-high sparsity workloads, making it ideal for sparse neural network acceleration.